

Voice Based Early Diabetes Diagnosis Using a Transformer Model with Intelligent Chat Support

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Declaration

We declare that the project work “**Voice-Based Early Diabetes Diagnosis Using a Transformer Model With Intelligent Chat Support**” submitted in the department of Electronics and Communication Engineering (ECE), Chaitanya Bharathi Institute of Technology, Hyderabad in fulfillment of degree for the award of Bachelor of Engineering is a bonafide work done by us, which was carried out under the supervision of Dr. T. Aravinda Babu

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Abstract

The detection of diabetes is of prime importance to avoid complications and to provide timely treatment. The existing techniques for the detection of diabetes, such as blood tests, clinical analysis, etc., are invasive in nature and not suitable for continuous monitoring. In the last few years, various machine learning techniques such as SVM, Decision Trees, CNN, etc., have been studied for the prediction of diabetes. However, the techniques have limitations as they need structured data and do not include the temporal relationships in the sequential data such as speech. This paper proposes a non-invasive method for diabetes prediction using the transformer model based on deep learning, where the model will analyze the features of the input voice signals and classify the individuals into diabetic or non-diabetic using the self-attention mechanism. Compared to the conventional methods using the voice signals for diabetes prediction, the method will show improved accuracy and reliability in the analysis due to the handling of long-range relationships in sequential data. The main contributions of this paper include the development of an adaptive learning mechanism where the model will be updated continuously using new voice samples during the testing phase, thus enhancing the reliability of the model. The method will also be integrated with an intelligent chatbot where we are analysing different level of understanding from the proposed algorithm and existing algorithms analyse the different parameters to come under diabetic cases, thus enhancing the overall interaction. The results obtained from the experiment using the transformer model show improved accuracy, compared to the conventional methods such as SVM, XGBoost, CNN, thus making the method more accurate, flexible, and interactive for diabetes prediction.

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List of Abbreviations

DL	–	Deep Learning
ML	–	Machine Learning
MFCC	–	Mel-Frequency Cepstral Coefficients
ANN	–	Artificial Neural Network
CNN	–	Convolutional Neural Network
SVM	–	Support Vector Machine
API	–	Application Programming Interface
NLP	–	Natural Language Processing
UI	–	User Interface
OS	–	Operating System
CPU	–	Central Processing Unit
GPU	–	Graphics Processing Unit
RAM	–	Random Access Memory
BMI	–	Body Mass Index
HbA1c	–	Glycated Hemoglobin
WAV	–	Waveform Audio File Format
Hz	–	Hertz
ReLU	–	Rectified Linear Unit
Softmax	–	Softmax Activation Function
Adam	–	Adaptive Moment Estimation
Django	–	Python Web Framework
PyTorch	–	Deep Learning Framework
Librosa	–	Audio Processing Library

CHAPTER – 1

INTRODUCTION

1.1. Introduction

Indeed, technology improvements have enabled the development of new ways of analyzing the healthcare industry. In terms of health care problems affecting people, diabetes mellitus is one of those illnesses which has caused serious concerns to healthcare practitioners and researchers. This disease is characterized by high levels of sugar in the blood because the body fails to utilize insulin effectively.

According to the latest global statistics on the health care sector, there has been a rise in the population of people suffering from this illness. Some of the main causes of this trend include unhealthy lifestyles, obesity, stress, and genetics. It can be seen that most diabetic patients do not receive diagnosis until it is already too late. On the contrary, their conditions become worse and expose them to the risk of developing other illnesses.

Conventional methods for diabetes detection include fasting blood glucose tests, oral glucose tolerance tests, and HbA1c analysis. Although these methods provide accurate results, they are invasive in nature as they require blood samples. In addition, these tests depend on laboratory facilities, trained medical professionals, and time for processing, which makes them less suitable for frequent monitoring. Many individuals avoid regular screening due to discomfort, cost, or lack of accessibility, especially in rural and underdeveloped areas.

There has been growing interest in developing automated and non-invasive means of disease detection due to advances in artificial intelligence technology in recent times. Machine learning models have gained great popularity in healthcare due to their potential to predict diseases based on data provided to the system. Many machine learning techniques including SVM, Decision Trees, Random Forests, and ANN have been proposed for predicting diabetes using structured data. This type of data includes features such as blood glucose level, BMI, age, insulin level, and blood pressure.

However, even though these techniques prove efficient, there are certain limitations to

their application. Firstly, the majority of these systems are developed on the basis of numeric data received through clinical trials and, therefore, cannot be applied without invasive testing. Secondly, none of these approaches is capable of processing non-numerical data, which includes both physiological and behavioral parameters of patients, such as voice signals. Therefore, there is a need to develop new technologies that would allow making use of other data sources to diagnose diseases.

Recent research shows that human voice can be regarded as a marker of various diseases. The production of voice depends on numerous physiological factors, including the volume of lungs, vocal cord activity, and nervous system functioning. The modification of these factors due to certain disorders, such as diabetes, leads to changes in voice parameters, namely pitch, tone color, frequency, and articulation. The alterations are often not noticeable to the human ear, but can be traced with the help of machine learning algorithms.

However, there are various advantages associated with the voice-based approach for disease diagnosis compared to other approaches. First, there are no invasive procedures involved in voice-based diagnosis, thus making it a safe approach that does not cause any discomfort to the patients. Second, the voice-based technique does not require any sophisticated medical tools. Instead, simple gadgets such as mobile phones and microphones are enough for the purpose. As a result, people can record their voices continuously without any difficulties. This makes it possible to detect disease and other health conditions early enough.

The present project focuses on developing a transformer-based deep learning approach to analyze the voice and make diabetic diagnoses. Transformer is the name given to a particular type of artificial neural network architecture that uses the self-attention model for processing sequential data. Unlike traditional models, including RNNs and CNNs, transformer-based models have become popular in recent years owing to their novel approach to learning from data.

The first phase is the collection of users' voices via the interface of the web application. Then, the voices are cleaned and normalized through voice preprocessing methods, which include normalizing, sampling, and removal of noise. Following that, feature extraction is done using Mel-Frequency Cepstral Coefficients (MFCC). The spectral characteristics of speech signals are expressed using the coefficients, making it suitable

for use in speech processing because the coefficient models how people hear speech.

Next, the features obtained are analyzed using the transformer neural network to determine any patterns indicative of diabetes. Transformer neural network refers to a neural network architecture for analyzing sequential data. Such a neural network learns complicated connections between various components of sequential data by processing data sequentially through multiple layers of self-attentive and feedforward neural networks. After obtaining patterns from the data, users are classified into three categories based on their risk levels: Low Risk, Medium Risk, and High Risk.

An important characteristic of the proposed system is the fact that the system is an adaptive one. In other words, the system can learn from experience and adjust itself based on learning from fresh data. This is vital to make sure that the proposed system is able to adapt itself according to different inputs provided by users along with different speech patterns. Adaptation based on new data ensures that the proposed system is both accurate and robust enough.

Apart from the fact that the proposed system will have a predictive approach to dealing with the patients, there will also be a smart chatbot incorporated into the system which will provide patients with information regarding their health in relation to diabetes. Any questions patients might have related to their health issues would be answered and patients would also be given useful advice on how to maintain good health and avoid this disease.

The entire system has been designed as a web application utilizing the power of the Django framework to help ensure successful interaction between the client-side and the server-side. Users may upload or record audio data through the use of their browsers while also viewing the predictions and chatting with the chatbots instantly. The user interface of the system has been made as user-friendly as possible in order to cater even to users with limited technological expertise.

There are many benefits of the proposed solution when compared to other solutions that already exist. It is not only non-invasive and cost-effective but is also very convenient for the users. Moreover, there will be no need for people to visit labs for testing purposes as they can simply get themselves tested at home.

Concerning applicability, this project finds application in various fields such as individual health surveillance, remote patient surveillance, and telehealth. In addition,

it could help individuals self-assess, support health experts in carrying out initial screenings on patients, and create awareness about diabetes among poor and rural communities. This particular project also gives opportunities to expand it into a diagnostic tool for other illnesses by following a similar approach.

In conclusion, this project demonstrates the potential of combining voice analysis, deep learning utilizing transformer technology, and artificial intelligence-based chatbots in coming up with an innovative tool for detecting diabetes at an earlier stage. Previous models lack certain qualities that differentiate this project from others. There have been improvements made in technology, which is advantageous for the medical sector as well.

1.2. Aim and Objectives

1.2.1 Aim

To design and develop a non-invasive diabetes detection system using voice signals and a transformer-based deep learning model, integrated with an intelligent chatbot for user interaction.

1.2.2 Objectives of The Project

- To develop a deep learning based prediction model for diabetes
- To design a non-invasive diabetes detection system based on voice through Transformer model
- To create a user interface and integrate it with a chatbot for diabetes related user support
- To design a accurate model by comparing voice based model performance and its results with the existing solutions

1.3 Motivation of The Work

There is no dearth of motivations in this project, considering the rising prevalence of diabetics and the lack of efficacy of currently available methods of diagnosis. People tend to avoid visiting doctors owing to the inconvenience, costs, and hassles associated with it.

Conventional methods used in diagnosing diabetics have a high dependence on

structured data, thus necessitating lab tests to achieve accuracy. These methods are not effective in tracking or screening diabetics living in impoverished areas.

There is, therefore, a need to develop a technology that will assist in detecting and ensuring that the individuals have awareness regarding their diabetes condition as early as possible. Such technology should be voice-based since the users are able to give responses in a convenient manner and without undergoing any medical tests.

Another reason why there is a need to develop this technology is due to the issue of user engagement. The technologies that have been developed currently only predict without assisting the user in any way.

Furthermore, the use of transformer architecture enables the improved accuracy of the system when dealing with sequential information such as audio data. Hence, this makes the system more reliable compared to traditional methods.

In conclusion, this research project seeks to bridge the gap existing between the health sector and the technology industry by developing an intelligent and user-friendly diabetes diagnosis app.

1.4 Scope of The Work

The first major scope of the current project shall cover the development of a smart and non-intrusive system that will be utilized in detecting any early signs of diabetes through voice sample data inputs. The main aim of this project will entail taking advantage of the latest developments in the field of artificial intelligence technology in developing applications related to health which can be easily used.

The main idea behind this project is based on using voices as an alternative input of data for medical uses. Therefore, the system currently being developed will be able to obtain voice samples through microphones or recorded voice samples which will be used as the inputs into the system.

Signal processing will employ sophisticated methods like normalization, noise elimination, and resampling of the input signals. Input features are extracted by using Mel Frequency Cepstral Coefficients (MFCC). Features extracted from MFCC include both the frequency domain and human perception of speech.

An essential feature of the proposed project is the use of the Deep Learning technique

using transformer-based neural networks for the analysis of the extracted features. The neural network proposed will be capable of analyzing sequential data and categorizing the patient based on their level of risk. Patients are grouped into three categories: Low-Risk, Medium-Risk, and High-Risk. This helps in the interpretation of the results by users who do not have any special technical expertise.

Furthermore, the system has a mechanism known as adaptive learning, where the machine is able to improve its effectiveness through acquiring knowledge from new data. Adaptive learning aids in increasing the accuracy of the machine in order for it to be more effective when put to use in the actual world because of the changes that could occur in the data.

Another crucial component of the system entails the creation of an online interface using Django. This interface enables users to interact with the machine through uploading their voice recordings as well as receiving predictions from the machine among others. The interface is simple to use even for those who lack the basic knowledge about machines.

Not only does the model predict, but it also comprises a chatbot that increases interaction with users. Users can ask questions related to diabetes and receive immediate responses. It works both online and offline, ensuring continuous accessibility of the data. Thus, it helps raise awareness among individuals about diabetes.

Regarding the application of the model, it is useful for various purposes. For personal healthcare, it assists individuals in conducting self-assessment and evaluating their health status. Telemedicine can benefit from the model by using it to conduct preliminary diagnosis and treatment. Rural areas that lack adequate medical infrastructure can make use of it for diagnosis purposes.

In addition, the proposed system will act as the foundation for further system improvements. For example, the system can be used to incorporate the mobile application, implement cloud computing to save data, and implement the system using wearable devices. Besides, the scope of the system might be expanded to include other illnesses that may be discovered using the voice recognition technique.

It is necessary to note that the proposed system is not a tool for diagnosis; instead, it is just a tool which helps identify potential problems. In some cases, the system can help diagnose some illnesses at an early stage.

1.5 Constraints

1. Data Dependency

The effectiveness of the proposed model relies heavily on the data utilized during the learning process as it should contain varied types of voices belonging to individuals from diverse age brackets and genders. To put it differently, an inefficient database cannot possibly contain all the varieties of human voices due to the physiological differences, gender differences, and other aspects. It therefore implies that a functional database ought to be fairly extensive, which is difficult to accomplish.

2. Background Noise and Audio Quality

There will be some form of noise, distortion, or inconsistency in the recorded voice signals because of the nature of the surroundings. This is due to such factors as the quality of the equipment being used for the recording, the proximity of the microphone to the speaker, and the sources of other noise forms. This results in the occurrence of errors in the MFCC feature extraction process.

3. Computational Requirements

Transformers-based models are highly computationally intensive, which implies that their training and prediction processes consume a lot of computing power in terms of memory, computing capacity, and processing time. Although training such models requires high-performance computers, implementing the model on low-performance machines such as personal computers or even mobile phones may result in poor performance due to the large amount of processing power required.

4. Internet Dependency

The system includes a chatbot module that provides advanced responses using external APIs. These features require a stable internet connection for proper functioning. In scenarios where internet connectivity is poor or unavailable, the chatbot operates only in offline mode with limited functionality. This may reduce the effectiveness of user interaction and restrict access to dynamic information.

5. Variability in Voice Characteristics

It is very dynamic based on different factors like age, gender, language spoken, emotion, sickness, accent, and so forth. Short-term factors such as fatigue can also affect one's voice. This complicates training because one must be able to distinguish between normal voice variations and those that belong to sick people.

6. Limited Medical Validation

Although the app uses complex algorithms for predicting the risks linked with diabetes, it cannot be substituted for clinical diagnosis. The application relies on probabilistic methods of making predictions; therefore, its findings might either reflect or not reflect the patient's state. Furthermore, being unable to undergo clinical trials and become certified, the app is not suitable for professionals' usage.

7. Dataset Collection Challenges

Collecting labeled voice datasets for diabetes detection is a complex and time-consuming process. It requires proper annotation, ethical approvals, and collaboration with healthcare institutions. Additionally, maintaining data quality and consistency is difficult. Limited availability of such datasets restricts the training and testing of the model, affecting its overall accuracy and robustness.

8. Security and Privacy Concerns

It will be concerned with handling confidential information concerning the user, such as audio files containing personal information on individual persons. Any misuse of the information by an unauthorized party will raise concerns for the privacy of the users. Yet, securing the information by making use of different security approaches such as encryption is inevitable but adds to the complexity of the system.

9. Hardware and Accessibility Constraints

This technology needs certain hardware equipment like microphones, computers, and Internet connection to work. In places that do not have any technological advancement, one can have difficulties accessing the software. There may also be problems related to the hardware needed for the use of these technologies.

10. Model Generalization and Adaptability

Even though the model might work very well on training data, it may not be able to work very well on unseen data. This is due to the variation in the recording environment, differences in user behavior, and differences in language. The model needs constant updating and learning for its optimal functioning.

11. Real-Time Processing Limitations

Although the algorithm is supposed to deliver its predictions instantly, some delays can be expected owing to the complications that arise from the sheer enormity of the model. Some delays will be observed in the feature extraction stage, creation of the prediction model, and delivery of the response.

1.6 Problem Identification

The present day methods employed to detect the presence of diabetes include the employment of invasive methodologies like the use of blood tests which need to be carried out in a laboratory environment and cannot be performed regularly. People do not like going through such regular processes as it causes them pain and they do not want to spend money on the process. Furthermore, such methods are not available for all people.

Although there have been some attempts at employing voice analysis techniques in the detection of diabetes, the issue of inconsistency in voices and the inclusion of noises in the voice recording process still remains. Moreover, the existing systems lack the feature of interacting with the users in a real-time environment.

CHAPTER-2

LITERATURE SURVEY

2.1 Voice-Based Diabetes Diagnostic Systems

With the development of the first ever voice-based diagnosis systems, we observe yet another breakthrough in the area of digital healthcare technologies when non-intrusive and accessible solutions become ever more sought after. The human voice is not only a form of communication but also a physiological signal determined by many bodily functions including the respiration system, the vibrations of vocal cords, and neuromotor control. Therefore, any alteration caused by chronic disorders like diabetes can produce small changes in voice characteristics.

Since diabetes influences several physiological processes such as hydration, nervous control, muscle function, it might cause alterations in pitch stability, frequency modulation, intensity, articulation patterns, etc. Such differences, although hard for human recognition, can be detected via special computer software.

In another study carried out by Elb'ejji et al., the researchers have shown that it is possible to use the acoustic characteristics extracted from voice recordings for accurate prediction of Type 2 diabetes [1]. As it can be seen from their results, voice signals may act as reliable digital biomarkers, especially when working in combination with sophisticated machine learning techniques. At the same time, the work performed by Oreskovic et al. is devoted to the problem of predicting prediabetes based on classical machine learning models of voice data [2].

Additional progress has been achieved in implementing voice-based technologies in practice. The work performed by Summoogum et al. involves development of a conversational triage system where the virtual assistant takes an active part in collecting voice data from the client for immediate health evaluation [3]. This technology improves convenience for users as far as the communication between them and the system is conducted automatically. Ahmadli et al. developed an idea of estimating blood glucose level based on voice [5].

Moreover, Kaufman et al. demonstrated that the utilization of smartphone recorded voice samples to detect diabetes is efficient, thus, making the use of the technology highly convenient [7]. Additionally, Jeon et al. proved that the utilization of voice signals acquired from everyday life may function as a reliable measure of glucose levels [13]. The discussed research proves that there is a significant potential of using voice technologies for health purposes.

Nevertheless, there are some drawbacks that require more investigation. The voice signal has a very unstable character and may depend on such parameters as the person's language and accent, emotions, and the presence of environmental noise. Furthermore, the absence of standard data sets and approaches to the assessment of the effectiveness of developed models makes the process rather complicated. However, the ongoing research of the domain will positively affect the results in the nearest future.

2.2 Machine Learning Based Healthcare Systems

The application of machine learning technology has transformed the healthcare industry through the automation of medical data analysis and assistance in clinical decision-making processes. The use of machine learning algorithms for diabetes identification involves the classification of patients using structured databases of clinical parameters.

Various classical algorithms such as Logistic Regression, Support Vector Machines (SVM), Decision Tree, and K-Nearest Neighbors (KNN) have been implemented into predictive healthcare systems. Through the processing of relationships among input attributes and disease statuses, these models can produce accurate classification outcomes. Mahesh offered a review of classical machine learning algorithms in recognizing patterns and classifying information [15].

In addition, Jordan and Mitchell presented the wider applications of machine learning technology in diverse fields such as healthcare and highlighted the significance of machine learning in handling and analyzing large amounts of data [20].

Though effective in their own way, these models come with intrinsic weaknesses. They are largely dependent on structured data, which is usually collected through invasive methods like blood testing. As a result, their effectiveness is limited in situations where non-invasive data gathering is preferred. Moreover, standard machine learning models depend on handcrafted features, which might fail to identify all possible data trends.

Moreover, these models are unable to work with unstructured data such as speech data. Speech data includes temporal and spectral properties, which cannot be analyzed by ordinary algorithms without additional processing methods. Also, these models are less efficient in analyzing nonlinear interactions and high-dimensional correlations.

These drawbacks necessitate better solutions to deal with different types of data and analyze complex signals.

Machine learning has been instrumental in the development of analytics in healthcare, especially with respect to predicting models on structured clinical data. Classical machine learning models like Support Vector Machine (SVM), Random Forest, Logistic Regression, and K-Nearest Neighbor (KNN) have been extensively used in classifying diseases.

For instance, Swapna et al. established that machine learning models are capable of classifying diabetic and non-diabetic patients through structured health parameters, exhibiting high levels of accuracy [23]. In their study, they found out that proper selection and processing of features significantly improve the effectiveness of the machine learning models.

In another review, Kavakiotis et al. examined the application of machine learning and data mining in diabetes studies [24]. Based on their results, it is clear that even though classical machine learning models yield satisfactory results in structured datasets, they fail when dealing with unstructured datasets, for example, images, speech, and sensors.

In addition, they noted that clinical data are prone to imbalance, missing data, and variability issues, which hinder the generalization of the machine learning model.

2.3 Speech Analysis Using Deep Learning

In speech processing, the use of deep learning algorithms has brought about significant progress since the method allows extracting features automatically and modeling complex structures using unstructured data. Unlike classical machine learning techniques, the deep learning approach requires no manual preprocessing of the input data.

The Artificial Neural Network (ANN), Convolutional Neural Network (CNN), and Recurrent Neural Network (RNN) have been applied for speech recognition tasks. The

CNN algorithm is highly efficient in analyzing local patterns within the spectrogram, while RNN models are designed to work with sequences. Heaton emphasized the capability of deep learning approaches to discover complex relationships between variables within huge datasets, which makes the technology relevant to the healthcare sector [16].

On the other hand, RNN-based models face some difficulties when working with sequential data, including vanishing gradients and long-range dependency problems. Consequently, researchers have come up with new transformer models, which have become state-of-the-art for processing sequence data.

Transformers make use of self-attentions that allow them to process all elements in the sequence together at once instead of processing them one after another. They are thus able to identify not only short but also long-range dependencies and perform more efficiently.

It has been shown by Kaufman et al. that deep learning models can be used to identify patients with diabetes based on voice analysis, achieving greater accuracy than traditional approaches [7]. In addition, Jeon et al. provided evidence for using voice data as biomarkers for abnormal blood sugar levels [13].

As a result of applying transformers for speech analysis, numerous opportunities have emerged in the field of medicine since they offer increased accuracy, generalization ability, and robustness to noise.

The evolution of transformer-based architectures has brought a revolution to the domain of speech and language processing. The most prominent development in this field is the development of BERT [28] model by Devlin et al., based on transformer architecture and leveraging bidirectional self-attention. As compared to other sequential models like RNN and LSTM, BERT learns the context representation by looking at the whole sequence of input at once, enabling it to learn dependencies between all tokens present in the input sequence.

Such bidirectional understanding is highly beneficial in healthcare-oriented conversation systems since the context in conversational models is essential to accurately interpret user input. For applications that leverage speech input in the field of medicine, such transformer-based models can be useful in enhancing the classification performance by learning dependencies in the sequential signal.

Additionally, transformer architectures are efficient in dealing with long dependencies in the speech signal and hence are very useful in biomedical signal processing tasks.

Rabiner and Juang made pioneering contributions to speech recognition and signal processing [29]. They developed methods of statistical modeling, Hidden Markov Models (HMMs), and acoustic features, paving the way for today's speech processing systems. The use of MFCC extraction, windowing, and spectral analysis in contemporary deep learning approaches stems from their pioneering work.

Combining all these contributions creates the foundation for applying speech signal processing techniques together with deep learning algorithms in healthcare settings.

2.4 Non-Invasive Healthcare Monitoring Systems

In line with the growing interest in patient-centric healthcare technologies, non-invasive monitoring systems have been developed. Such systems intend to conduct constant health assessments while avoiding the use of invasive techniques to ensure greater patient satisfaction.

Non-invasive healthcare monitoring systems employ various data sources other than invasive methods. Examples include wearable sensors, voice signals, and physiological information which can be used to conduct a continuous evaluation of patients' health status. In particular, non-invasive methods can be used effectively to monitor the health of patients with chronic illnesses, including diabetes.

For example, Sidorova et al. considered the potential of using voice signals for estimating the blood glucose levels [9] and emphasized the potential role of voice signal in non-invasive healthcare monitoring. Additionally, this research explored the problem of dealing with the variability of voice data.

Furthermore, Pinyopodjanard et al. examined the acoustic properties of voices in diabetics and compared them to the ones observed in non-diabetics [11].

The non-invasive systems have a number of benefits, such as lower cost of treatment, better availability, and more comfortable patients' experiences. These types of technology are most helpful in underdeveloped places where there is not enough medical infrastructure available. The main problem with them is maintaining precision due to different kinds of inputs and settings.

Non-invasive health care monitoring devices have been gaining considerable relevance in modern medicine owing to their capacity of continuously assessing a patient's health without any discomfort. The purpose of this technology is to make the process of measuring blood parameters completely non-invasive.

For instance, Caduff et al. studied various non-invasive glucose monitoring methods and underlined their contribution to the field of diabetes treatment [25]. The paper discussed optical, biosensor, and signal measurement methods for glucose estimation as well as their difficulties of accurate calibration and environmental sensitivity.

Furthermore, Vashist considered non-invasive methods of glucose monitoring and divided them into analytical chemistry and sensing ones [26]. The study concluded that although there are advances in this area, the problem of obtaining clinically reliable results still exists since different physiological responses to changes affect accuracy.

The use of speech for health monitoring represents an innovative approach to this field. The voice contains both physiological and neurological data about a person that can be used as a digital biomarker for chronic disease detection, including diabetes. Nevertheless, the variability of voice signals based on the emotional state of the user and other factors remains a problem.

2.5 Hybrid AI-Based Healthcare Systems

Hybrid artificial intelligence solutions constitute one of the most sophisticated means of implementing artificial intelligence systems because they combine different approaches for improving the efficiency of computer programs. In this case, machine learning, deep learning, and rule-based models can be used simultaneously for benefiting from their advantages.

In particular, Scotti highlighted the significance of using artificial intelligence in implementing different decision-making algorithms for obtaining better results [14]. At that, hybrid healthcare systems usually incorporate predictive tools and interactive components like chatbots.

As chatbots enhance the interaction process through immediate communication between a system and its user, it is possible to assume that they help people comprehend their health status better, respond to their questions, and suggest necessary actions.

Thus, chatbots contribute to improving the involvement rate of users.

In contrast to standalone models, hybrid artificial intelligence-based systems are able to work with both structured and unstructured datasets. Moreover, they increase system reliability by utilizing several decision-making algorithms simultaneously. Nonetheless, their implementation necessitates additional efforts on the part of developers.

The idea behind hybrid AI models is that a combination of several computation techniques, including machine learning, deep learning, and rule-based reasoning, is the most promising trend in healthcare analytics.

Transformer-based models, presented by Devlin et al., brought breakthroughs in natural language comprehension due to the application of self-attention mechanisms [28]. Nowadays, transformers like BERT are considered the basis of most modern conversational AI solutions.

As for hybrid healthcare systems, the main concept is a combination of predictive models with the conversational part to get both the possibility to predict the result and explain it to users through natural language processing tools.

This combination will increase access to the system and its engagement from the perspective of limited technical knowledge among potential users. Nevertheless, developing an effective hybrid system is challenging since it implies balancing between computational complexity and performance.

2.6 Comparison Between Existing Methods

An analysis of the existing methods for detecting diabetes gives an understanding of how each of these technologies works, what their advantages and disadvantages are, etc. Traditional invasive techniques based on blood test have high accuracy, however, they are not convenient and quite time-consuming. Machine learning techniques are more accurate, but require the input data to be organized. Deep learning has better results compared to traditional machine learning methods, but requires larger amounts of data and computational power.

The most novel non-invasive approach consists of using voice characteristics for the detection of diabetes. This technique does not require special infrastructure and may be

considered highly flexible. However, voice characteristics may vary between individuals due to various factors; thus, voice detection algorithms may encounter certain challenges. Early studies conducted by Chitkara and Sharma have proven the feasibility of voice-based detection of diabetes [18]. The importance of diabetes detection has been emphasized by Kharroubi [19].

From the comparison above, it can be seen that there is no ideal method of diabetes detection available today. Thus, an integrative approach should be adopted.

The comparative review of current approaches toward detecting diabetes provides insight into the pros and cons of each method. Invasive procedures like blood glucose measurement deliver highly precise measurements but are unsuitable for regular monitoring because of their invasive nature.

Approaches based on machine learning deliver more efficient predictions based on structured data analysis but have inherent drawbacks of depending on manual feature extraction and the incapacity to process unstructured information.

In turn, deep learning models allow automatic feature extraction based on hierarchical analysis of data. Yet, deep learning algorithms demand considerable amounts of data and computational power for effective training.

The foundational works of Rabiner and Juang in speech processing [29] remain crucial in contemporary voice recognition systems through delivering techniques for feature extraction and acoustic modeling necessary for turning speech signals into valuable representations, including MFCCs.

Feasibility of voice recognition was already proved by Chitkara and Sharma when they showed that medical biomarkers can be contained within speech signals [18]. The problematics of increasing number of people suffering from diabetes around the world is described by Kharroubi [19].

Moreover, Schuller et al. highlighted the significance of speech analytics for health and the increasing relevance of such approaches in artificial intelligence applications in medicine [20]. Thus, the results confirm the high potential of speech recognition techniques combined with machine learning and deep learning for use in healthcare in the future.

In general, comparative analysis revealed that there is no single method adequate for

diabetes screening and monitoring. In this context, the use of hybrid technologies involving voice analytics, deep learning algorithms, and conversation AI is more promising.

CHAPTER-3

METHODOLOGY

3.1 Proposed System Overview and Process Framework

The proposed system would be an advanced and non-invasive technique for the early detection of diabetes by utilizing voice signal data as its primary input data. For this purpose, the techniques used would consist of signal processing techniques, transformer neural network machine learning techniques, and intelligent chatbot application for generating a total healthcare diagnosis system. It would be hoped that such a system could predict diabetes at an early stage, and more importantly, it should be easy to access instantly.

It should be noted that this proposed system's working mechanism would be based on the fact that the speaking process in humans involves the body physiology and metabolism of humans. Speaking would involve the activities of lungs, voice chords, and articulation apparatus as well as the neurological control of these functions. In case there is any disease like diabetes which affects these processes, it may result in the creation of slight deviations in speaking activity. These deviations can take place in terms of voice pitch, spectrum, volume, articulation, and timings.

Diabetes can also affect the status of hydration, sensory nerves, and muscle coordination, leading to changes in the modulation of the voice. Such physical effects on the voice modulation cannot be seen with our eyes but can be computed using certain algorithms to find any correlation between voice modulation and diabetes.

The pipeline for voice diagnosis is composed of several steps, starting from data acquisition to diagnosis itself. All of the steps involved in the pipeline are important since their results are instrumental in obtaining final predictions.

The intelligent chatbot is used in the framework not only for prediction but also for ensuring interaction between the user and the system. This is done via using the intelligent chatbot, which ensures provision of information by the system about the worries of the users and makes the system very practical.

On the whole, the above framework aims at solving the challenge of integrating artificial intelligence and health applications in order to come up with an effective approach to early diabetes detection.

UML Diagram

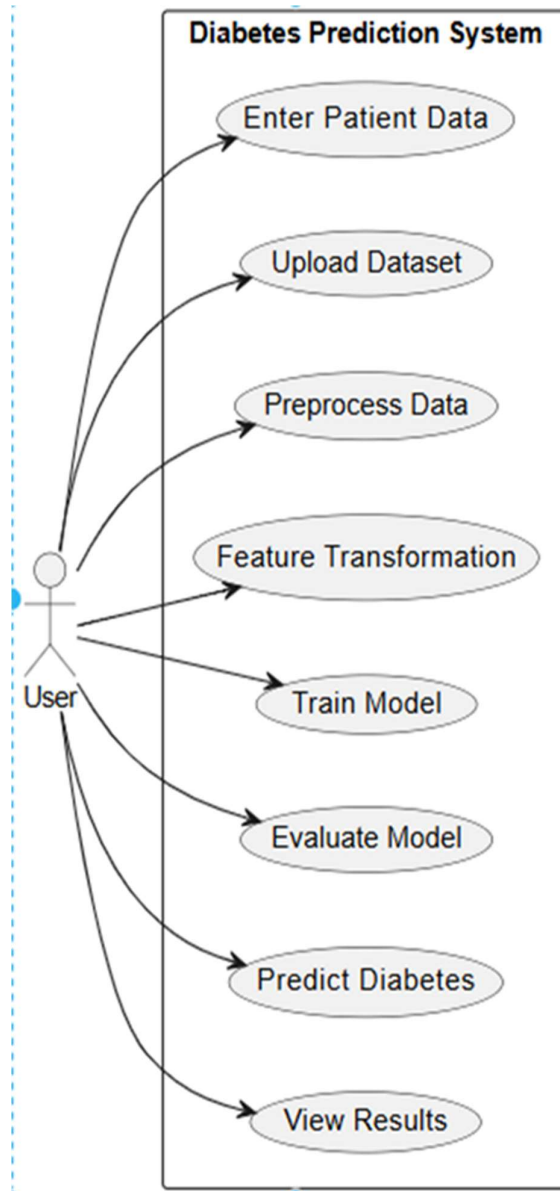


Fig 3.1 1 Use Diagram of Diabetes Prediction System

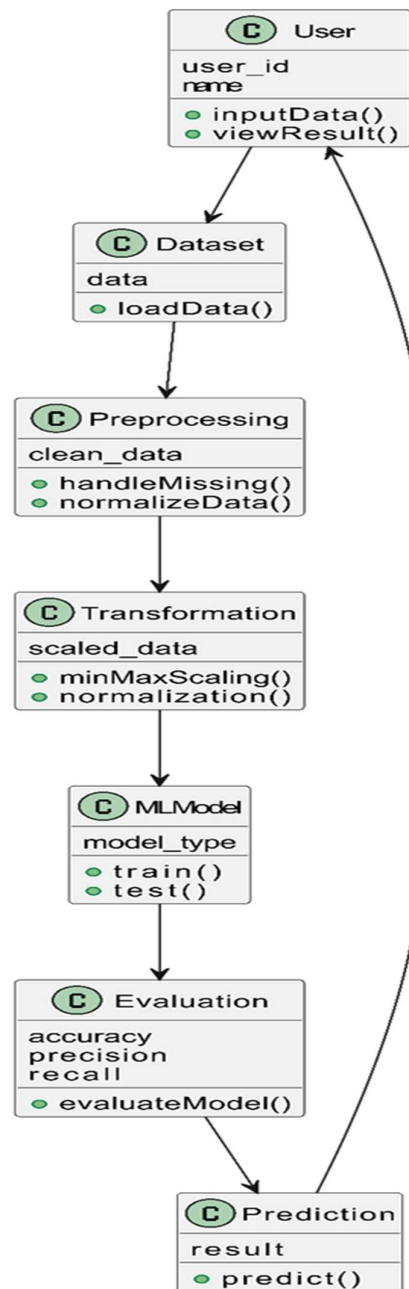


Fig 3.1 2 Class Diagram of Diabetes Prediction System

3.2 Overall System Workflow For Diabetes Detection

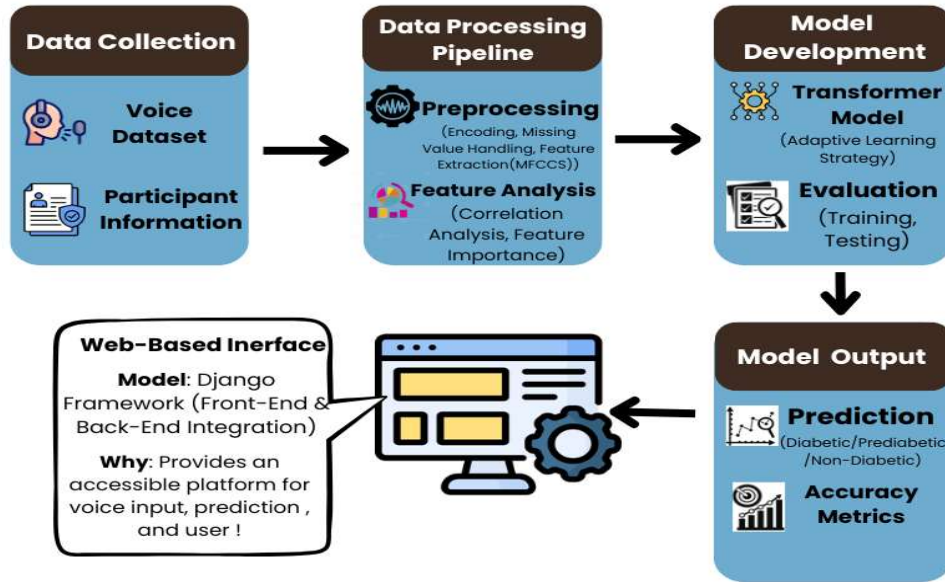


Fig 3.2 1 Workflow of Diabetes Detection System

The whole event process in the algorithm design process guarantees efficient processing and predictive outcomes with regards to diabetic risk prediction. First, we have the collection process of the voice inputs by the user of the internet platform. Such inputs will form the basis of our dataset in this project.

After the acquisition of audio inputs, we then proceed to the processing phase. The core goal of this phase is to reduce noise levels within the dataset. The phase is quite critical in our analysis since it helps to clean the acquired dataset.

The outcomes of the categorization process are presented to the user via the web interface. It guarantees that the results are delivered in a comprehensible manner. This method enables real-time analysis and provides an excellent user experience.

3.3 Acquisition of Voice Data and Signal Preprocessing

The process of voice sample data collection is performed at the stage of data acquisition. Users are provided with the freedom to record their voice in real-time mode or upload the recorded files on the computer.

It should be emphasized that in practical applications, the signals may contain distortions caused by background noise, microphone quality, and speech delivery characteristics of the user. Thus, preprocessing is an integral part of the application.

Preprocessing implies performing a number of operations, including resampling, noise reduction, and normalization of the signal. First, the process of resampling includes bringing the sampling rate to the desired value (22,050 Hz). It is then followed by noise reduction to improve the quality of the signal. Normalization helps to achieve uniformity in the input data and thus to prepare it for feature extraction.

The feature extraction technique adopted in the proposed algorithm involves the use of Mel Frequency Cepstral Coefficients (MFCC). This method is popularly used in voice recognition because it models the human hearing system. The audio signals are converted into a series of coefficients that model their frequency spectrum.

The MFCC representation is expressed as:

$$\text{Equation 1 : } MFCC(n) = \sum \log(S_k) \cos [\pi n(k - 0.5) / K]$$

where S_k represents the energy of the Mel-scale filter banks.

In this system, **40 MFCC coefficients** are extracted for each frame, forming a feature matrix:

$$\text{Equation 2 : } X \in \mathbb{R}^{(T \times 40)}$$

where T represents the temporal length of the signal. Padding is applied to ensure uniform input size for the model.

Table 1: Audio Preprocessing Parameters

Parameter	Value
Sampling Rate	22050 Hz
MFCC Coefficients	40
Audio Format	.wav
Padding	Applied

3.4 Feature Learning Mechanism Based on Transformer Network

The transformer network-based deep learning model acts as the analytical engine for the proposed approach. It has been developed for handling sequential datasets like speech signals and finding relevant features related to diabetes.

As opposed to conventional models like RNN and CNN models, transformers are not confined to sequential analysis. Transformers implement a self-attention mechanism that helps the model analyze connections between elements in the dataset regardless of their position. This way, the transformer can learn short and long-term dependencies in the speech signal.

First, the processed MFCC feature matrix is analyzed by the embedding layer to convert input features into a high-dimensional space:

$$\text{Equation 3 : } Z = XW + b$$

Such mapping helps boost the expressiveness of the features in order to allow learning of more sophisticated relationships. Thus, the embedding layer helps project 40-dimensional MFCC features to another (richer) feature space (e.g., 128 dimensions).

Subsequently, the embedded features are fed into the encoder of the transformer architecture consisting of multiple layers applying multi-head self-attention and fully connected feed-forward networks. Self-attention operation is implemented as follows:

$$\text{Equation 4 : } \text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) V$$

Where Q, K, and V are query, key, and value matrices obtained from the sequence of the input vectors. Self-attention mechanism calculates the level of association between elements of the sequence with one another in order to focus on the most important features.

One of the main strengths of the proposed method is an efficient way of handling long-range dependencies in speech signals. Such dependencies mean that important features of speech can be located in other time intervals, and therefore self-attention makes it possible to detect such relationships.

The encoder layer is repeated several times, and then the output representation is pooled using the following mean-pooling operation:

$$\text{Equation 5 : } Z_{\text{avg}} = (1/T) \sum Z_t$$

This is the vector that embodies all the features of the speech signal, and it is applied to classify.

In summary, the transformer algorithm serves as a robust means of analyzing the voice signal, allowing for the effective identification of patterns related to diabetes.

3.5 Diabetes Risk Classification Strategy

Classification refers to transforming the result of transformer model output into useful and understandable information. This phase ensures that the system produces useful and actionable insights.

The feature vector that is the result of the transformer model is fed into a fully connected (dense) layer which finally classifies the data. In this layer, the result is a vector of probabilities per class:

$$\text{Equation 6 : } y = \text{softmax}(W_o Z_{avg} + b_o)$$

The output of this layer is normalized with softmax such that all the probability values sum up to one. The probability value denotes the possibility of the input being classified in a particular class.

Finally, the argmax operation is used to determine the final predicted label:

$$\text{Equation 7 : } \hat{y} = \text{argmax}(y)$$

The classification result determines the risk level of the user as follows:

Low-risk: Denotes that the voice characteristics have little sign of diabetes.

Medium-risk: Shows that the individual is at risk of diabetes hence they should take note of their health.

High-risk: Denotes that the person is highly probable to be diabetic.

The aim of this method of categorization is to make the analysis of the output simple for the user. The user will not have to deal with complicated numbers but rather simple categories.

Furthermore, the system could be made more complex by incorporating confidence values and health advice.

3.6 Intelligent Query Processing and Chatbot Workflow

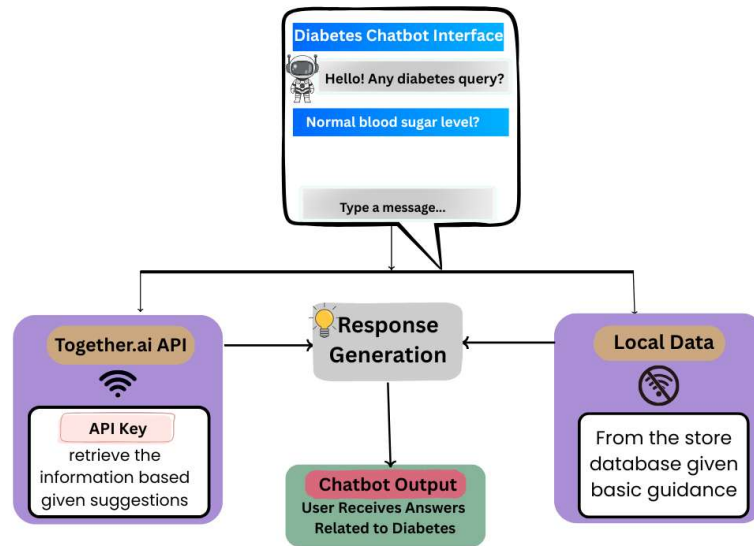


Fig 3.6 1 Chatbot Interaction Workflow

The chatbot module is an important component of the software solution, which was created in order to improve user interaction and help them in addition to the forecasts made. The virtual assistant that will be able to answer the questions of the users on diabetes, its symptoms, prevention, and other guidelines will be the chatbot.

The chatbot implementation process starts with the entry of the question made by the user into the web application. Next, natural language processing methods are used for the text entered, like tokenizing, normalization, and removing unnecessary characters.

Later on, the keywords extraction and intent recognition steps will be used in order to determine the purpose of the user's request. The response generation process will then depend on the user's purpose.

Some conditions to control the process of response generation are as follows:

The following actions would take place if there is a valid API key:

- The application will connect to an external language API server in order to generate responses.
- If there is no valid API key, then it will depend on the existing set of rules and the knowledge base which is based on FAQs related to diabetes.

Combining both approaches ensures that the chatbot is always available under any

circumstances. Even when API assistance is not available, the solution is capable of providing accurate responses.

Some advantages associated with integrating the chatbot into the system include:

- Timely recommendations
- Increased user interaction
- Enhanced understanding of diabetes
- Decision-making support

3.7 Adaptive Model Learning and Performance Optimization

In order to make the algorithm more efficient, it is important to come up with a way of dynamic learning where the algorithm will become better and better with time through learning.

The model is trained through various numbers of epochs using voice data sets. Performance of the model is assessed through various metrics such as accuracy and loss in every epoch.

Another key characteristic of the adaptive algorithm is generalization. Not only does it have to perform well on already seen examples, but it must be able to cope with unseen examples. Such concepts as regularization and validation prevent the occurrence of overfitting.

Moreover, the system adopts an incremental learning method, which makes it possible to introduce new information into the system without restarting the whole learning process.

Performance optimization is also achieved through:

- Efficient feature representation (MFCC)
- Transformer-based architecture
- Optimized hyperparameters (learning rate, batch size)

These techniques ensure that the system maintains high accuracy while minimizing computational overhead.

3.8 Web-Based System Integration and Data Flow

The creation of this system is supposed to be carried out through the web application

which involves the front-end and back-end aspects so that the system can run smoothly. This aspect aims at making sure that the solution is always accessible, scalable, and interactive in nature.

In this case, the front-end aspect plays an integral part in interacting with the users, thus allowing them to upload voice samples and predictions along with interaction with the chatbot.

All computations are carried out by the backend, which is built on the Django framework. The computations will include the following processes:

- Pre-processing of audio information

- Extraction of features

- Inferences

- Generation of chatbot responses

There is always HTTP requests that allow connecting between the frontend and the backend to make an effective transfer of data between these two systems. When users send their audio information, this information will get to the backend for computation and analysis purposes before being sent to the frontend.

The application is able to provide its computational capabilities in real time, offering users immediate responses. Real-time processing is enabled due to efficient handling of data and optimization of machine learning models.

Thus, frontend, backend, and voice analytics work as one system allowing creating a convenient solution for diagnosing diabetes without any invasions.

CHAPTER 4

IMPLEMENTATION AND RESULTS ANALYSIS

4.1 System Implementation Overview

In addition, questionnaire-based systems are also used to estimate diabetes risk by collecting user input related to lifestyle and medical history. However, these approaches are highly dependent on user-provided data, which may be incomplete, inaccurate, or biased. As a result, the reliability of such systems is often compromised.

Another major limitation of existing systems is their inability to effectively process sequential data. Speech signals contain temporal dependencies that cannot be captured by conventional machine learning models. Even deep learning models such as convolutional neural networks, which are capable of handling signal data, often struggle to capture long-range dependencies in sequential inputs.

The system is developed on the basis of the Python programming language as the main one due to the availability of numerous libraries and frameworks that are needed for developing machine learning algorithms, audio signal processing, and web applications. The flexibility of the language, easy code implementation, and high popularity in such kinds of projects made it possible to develop the project quickly. An audio signal can be processed with the help of Librosa, while PyTorch is suitable for implementing transformer-based deep learning models.

For working as the control center in the system, handling user requests and data flow between components, the Django framework is used in the backend. The MVT architecture uses the Django framework and consists of three components – Model, View, and Template.

The front-end part of the software application is developed using HTML, CSS, and JavaScript languages. It provides the ability to develop an easy-to-use user interface through which clients can interact with the system. Because of the friendly user interface, the users of this software application can use it efficiently, regardless of their technical background. This software allows them to upload their voice recordings and receive predictions by interacting with the chatbot.

This software operates on the concept of the modular design pattern. The software consists of various modules that carry out different operations within the software application structure. Using the modular architecture approach allows the developers to update specific parts of the software without interrupting its operation. Key modules used within this software application include:

The input module – this module collects the voice data entered by the users via their voice recordings or files.

Audio preprocessing module – it enhances the quality of the voice by eliminating all the noises from it and normalizing the amplitudes.

Module for Feature Extraction: Converts the processed audio into MFCC features that will represent the spectral properties of speech.

Inference Module for the Model: Employs a deep learning model based on transformers to process the generated features and yield prediction results.

Chatbot Module: Handles user requests and gives answers according to the authentication of the API key or knowledge base.

Module for Output: Presents the predictions and chatbot responses to the users through the web application interface.

The communication among these components is very efficient because of adequate data handling. Once the user provides their input data, the latter is transmitted to the backend where the feature extraction is performed. The next step includes sending the obtained features to the deep learning model, generating predictions.

One of the crucial elements of the implementation process is the real-time processing of information. In other words, users can receive instant feedback while working with the system. This is made possible through effective data flow and model execution procedures. It is worth mentioning that the system was created so that there would be no delay in getting results while maintaining high-level accuracy.

Firstly, it should be noted the introduction of the chatbot module, which will help in providing a more effective communication between the users and the system. The chatbot operates simultaneously with the process of predictions. That is why it is possible to get more information about treatment of diabetes. There is a possibility to provide information through the API; however, there is always an alternative function.

There are several basic functions for security within the system, which will help users stay secure. These include authentication, authorization, and sessions management.

As a whole, the design manages to integrate several technologies into one coherent system that can effectively diagnose diabetes without any invasive procedure using sound waves. The efficiency and scalability of the system have been enhanced by the modularity capability, real-time processing, and interactivity features. The design demonstrates the feasibility of integrating AI and Web technologies in healthcare.

4.2 System Setup and Server Execution

The creation of systems and starting of the server occurs in the implementation stage whereby the system will be created in terms of software development to implement the system.

The first process which should be done in the implementation stage is installing python programming language and other necessary libraries including:

Django, librosa, numpy, and pytorch. Django will be useful for developing web applications, librosa will be useful for processing audio signals. Numpy will be useful for carrying out numerical calculations while pytorch will be useful in developing deep learning algorithms.

The next process after installing the aforementioned libraries will be the process of designing the application according to the django framework by organizing the application into different components like Models, Views, Templates, and static files.

The last process for launching the server involves the following command line code as follows:

```
python manage.py runserver
```

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\dasar\OneDrive\Desktop\Diabetes voice based\Diabetes voice based>venv\scripts\activate
(venv) C:\Users\dasar\OneDrive\Desktop\Diabetes voice based\Diabetes voice based>cd backend
(venv) C:\Users\dasar\OneDrive\Desktop\Diabetes voice based\Diabetes voice based\backend>cd voice_diabetes
(venv) C:\Users\dasar\OneDrive\Desktop\Diabetes voice based\Diabetes voice based\backend\voice_diabetes>python manage.py
runserver
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).
March 17, 2026 - 20:34:15
Django version 5.2.11, using settings 'voice_diabetes.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.

WARNING: This is a development server. Do not use it in a production setting. Use a production WSGI or ASGI server instead.
For more information on production servers see: https://docs.djangoproject.com/en/5.2/howto/deployment/
```

Fig 4.2 1 Command prompt showing IP address and server execution

4.3 Web Application Access and Deployment

Web Application Access & Deployment includes the stage that comes after interaction with the system in order to confirm its functionality. Through web application access & deployment, the developed system can now be made operational in any operation system because of access through the web browser.

After development of the software, it is easy to deploy the software in any system provided the software works properly on the server. By default, the URL address of the system would be <http://127.0.0.1:8000/>. Therefore, the system can only work on itself. In order to make use of the system, it is important to use the IP address of the computer.

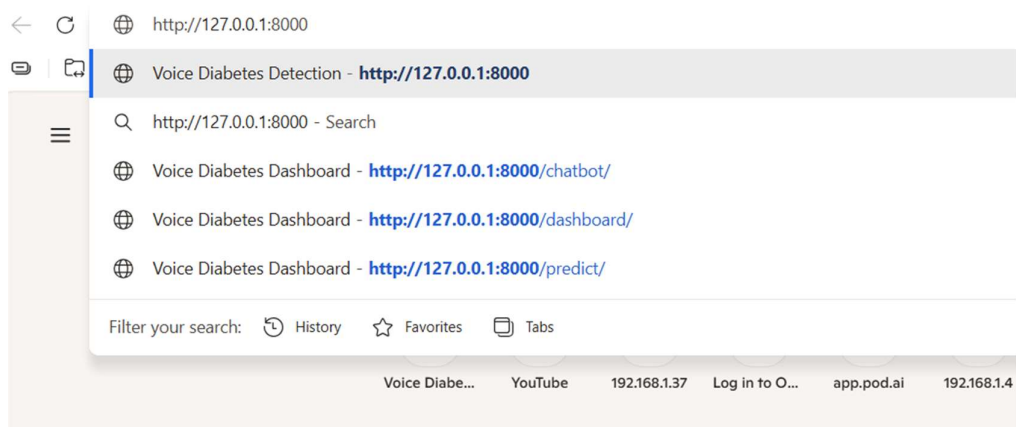


Fig 4.3 1 Accessing the system through local IP address

The application can be reached by using the IP address and server port using the browser's address bar in the following manner:

`http://<IP Address>:8000/`

Once the URL is accessed, the browser will generate an HTTP request to the Django server, which will reply to the request by sending back the required web page to render it within the browser.

There are numerous advantages in adopting the web application strategy to develop this application. For starters, it is not necessary for users to install any additional software for using the application. Users will be able to access the application from any device that has a web browser pre-installed.

In the deployment stage, all static and dynamic assets required for the proper functioning of the application will be configured. Static assets comprise JavaScript and CSS files that will improve the design of the application. Templates contain the structure of the web pages.

It is supposed to be used with many users at a time. The system uses the request-response model for handling user queries. Hence, it enables concurrent usage by many users. It is an advantage when using the system to showcase or evaluate the product.

Moreover, another key requirement in the application deployment is the fast loading and quick response. The application is quickly loaded into the browser and promptly responds to user inputs. To achieve this requirement, the optimization of the front end and minimizing data transfers between the client and server was performed.

Furthermore, the system has error handling capabilities. In case of any error in processing the requests from the users, the system will notify the users. The function is essential for preventing system crashes because of user mistakes.

Moreover, the deployment phase presents many possibilities of making the system scalable. Although the current implementation uses the local server in developing and testing the application, it can be expanded further to leverage cloud technologies to deploy it online.

Lastly, security issues were tackled during the application deployment phase. The sessions management was optimized to secure connections, and authentication mechanisms were applied to prevent unauthorized access.

To sum up, it is important to emphasize the importance of the stage of application access and deployment since without this phase the whole process would not be effective or efficient. This model becomes operational once converted into a web application where interaction with the application and obtaining predictions can be done.

4.4 User Authentication and Access Control

Herein lies the fact that the process of authentication and access control is considered to be the main component of application system design, since it provides opportunities to control the degree of security of access to the web application. As far as application development that is designed to meet healthcare needs is concerned, it should be mentioned that it is required to take into consideration the fact that personal data of the users will be treated as confidential. Thus, authentication process should be incorporated into the application system design.

It comes to the point that with respect to the method proposed to develop a web app, the authentication process is carried out by means of implementing registration, authentication and sessions management processes. Sessions management will be implemented by applying Django framework. That is, before accessing application services, the user will be expected to register by providing such details as a username, an email address and password.

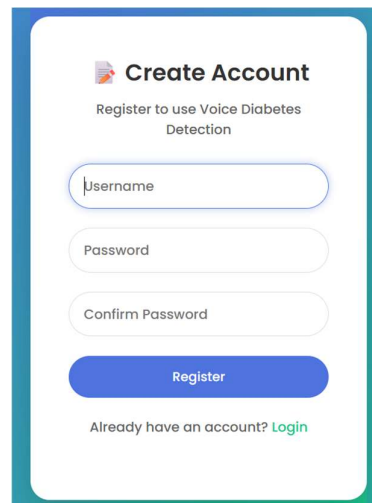
The image shows a user registration interface titled "Create Account". Below the title is a subtitle "Register to use Voice Diabetes Detection". The form contains three input fields: "Username", "Password", and "Confirm Password". Below these fields is a blue "Register" button. At the bottom, there is a link that says "Already have an account? Login". The entire form is enclosed in a blue border.

Fig 4.4 1 User registration interface

These details should be verified in order to ensure that all requirements are met. For example, the user should have only one username and the minimum number of

characters for password should be equal to eight.

This is a method used to ensure that the system recognizes the user accessing the system. The system then checks to see whether the details that have been keyed in are the same as the ones contained within the database. In case the verification does not take place, the user is allowed access, otherwise an error message will be displayed.

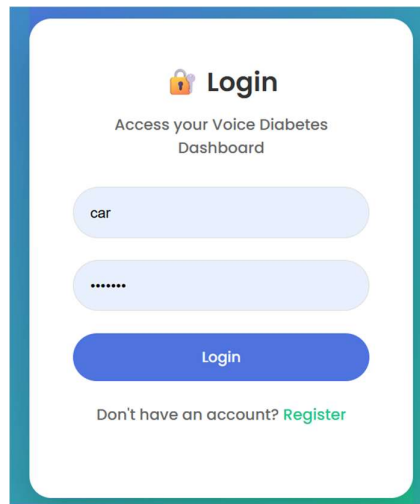


Fig 4.4 2 User login interface

There are session management approaches used in order to keep user sessions secure. After authentication, a session is created for the specific user. It will allow the application to identify the authenticated user through other pages and avoid authentication in future processes. This will not only help in improving the convenience of the user but also secure it as well.

In the application, there are certain access control measures that will be put into place in order to ensure that the user performs tasks that they have access to. For instance, prediction results and chatbot conversation are restricted to only those individuals who are successfully logged in.

The application ensures security standards through the use of some security techniques such as hashing passwords and encryption. This will make sure that there is no leakage of sensitive data in plain text form. The Django Authentication System that comes by default is very reliable in terms of security.

The problem of the user experience is yet another consideration that should be taken into account in the development of this authentication system. Amongst the goals set

during the design of this authentication system, there were some plans to create a system in which people would easily be able to register, log in, and use the authentication system properly.

To make sure that the software is user-friendly, procedures have been created to deal with possible errors. For example, if the login credentials cause some trouble for the user, he/she will be notified through the appropriate messages. In addition, the system will send notification to the user about any problems with the validation of data submitted for registration.

Among other security issues that should be included into the future development of this project there are two-factor authentication, some password recovery procedure, and sending emails. This would definitely enhance security in the process of authentication. As can be seen from the above discussion, the module has many positive qualities.

4.5 System Dashboard and User Interface

In this particular scenario, it is this interface that will act as the mechanism that will help the user in accessing the voice diabetes detector. Hence, the design of this interface will have a very crucial role in ensuring that the system remains user-friendly irrespective of the level of technological expertise of the person using it.

Once the user has logged into the system, he or she will directly find himself/herself on the dashboard from where all the features of the system can be accessed. The presence of this dashboard simplifies the process for the user.

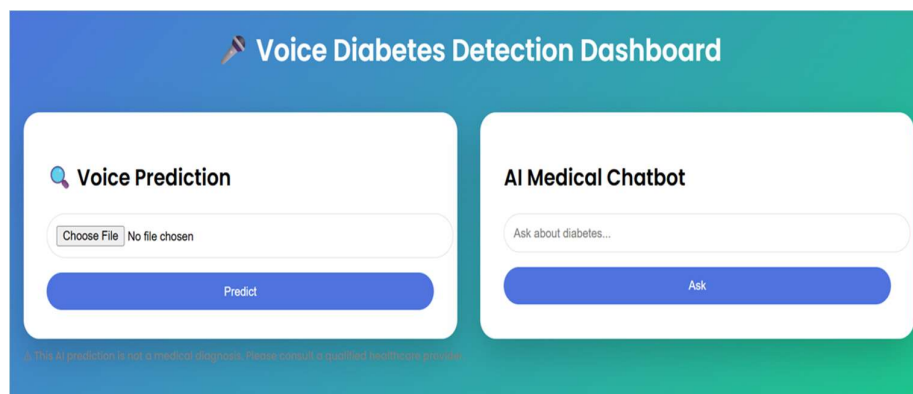


Fig 4.5 1 Dashboard interface of the system

Dashboard has been categorized in several manners for different purposes:

Voice Input Category:

In this category, the user may input his voice in the procedure. This is possible either through talking into the microphone fitted in his device to record his voice or by uploading the recording of his voice that he has already made. Instructions on how to conduct these procedures have been provided accordingly.

Prediction Output Category:

After inputting his voice, the input will be analyzed, and predictions will be made and shown to the user in the Prediction Output Category. These predictions have been classified into three categories which include; Low Risk, Medium Risk, and High Risk. Besides predictions, there may also be recommendations for the user concerning his condition.

Chatbot Category:

This section provides an opportunity for the user to ask any question to the Chatbot concerning his diabetic condition.

Control Elements:

There will be many control elements such as buttons and menus, which help the user to accomplish different activities, including uploading files and clearing inputs, among others.

HTML, CSS, and JavaScript are some of the primary tools which are utilized in the designing of the user interface. This allows the interface to be flexible across various different platforms.

In terms of usability, the dashboard has ensured the following features:

Simplicity: The user interface is designed in such a way that there is nothing confusing for the users in it.

Flexibility: The flexibility of the interface design ensures that it is adaptable to various screen sizes, from computer screen to even the screen of smartphones.

Clarity: The information which is important to users, like the results of predictions, is clear in the interface design.

Feedback: The users get feedback about the acceptance of the data they have provided to the interface.

Interaction with the user in real time can be another factor to be considered while developing the application. After typing a voice command into the interface, it immediately gives a response without requiring the users to manually reload the web page.

In addition, error detection is provided in the interface. When the file upload process does not go well or any user inputs are not recognized, the users get notified.

It is essential to underline that the interface plays the role of mediator between the front end and the back end. Each time when the users communicate with the interface, they create requests for the back-end server that gives its answers.

Concerning the dashboard, there are various enhancements that may be introduced in the following versions of the application. These include graphical presentation of results, personal user accounts, tracking of data history, and other options.

In summary, the design of the system dashboard and interface has been done in a way that makes it possible for the system to provide a seamless interaction for its users due to the integration of all operations into one system for easy interaction and communication of outcomes. Graphical presentation of results, personal user accounts, tracking of data history, and other options.

4.6 Audio Processing And Feature Extraction Results

Among the modules which have been developed for feature extraction from the input audio dataset, one of the modules which stands out as the most important is that for feature extraction of audio data. This is because the audio datasets generated from sensors are very poorly structured, containing noise, silence, and variance in the manner that people speak. Therefore, the need arises for an important module which can transform the unstructured audio datasets into structured features datasets.

4.6.1 Audio Signal Acquisition and Input Handling

The system acquires voice input through either one of the following mechanisms:

Real-time recording via microphone connection

File uploading of pre-recorded voice input

Upon capturing the voice input, it stores it as waveforms, wherein waveform indicates

the fluctuations of signal amplitudes with time as the building blocks of a speech signal.

The system is designed to ensure standardization of all audio inputs based on the following aspects:

Sampling rates of all signals should be kept constant

Consistency in audio format (such as the use of WAV formats)

Conversion to mono channel format in all cases

Such standardization is essential for achieving consistent results in extracting features.

4.6.2 Audio Preprocessing & Signal Conditioning

Before the actual feature extraction from the signal, there is an important stage when the voice is preprocessed in order to ensure its quality. Such a stage becomes necessary in order to remove all kinds of interference that can affect the results of the analysis.

The preprocessing involves the following steps:

Noise Reduction: Digital filtering of background noises allows keeping only those components of the signal which relate to speech itself.

Silence Removal: Removal of all non-speech fragments like silences and pauses from the voice sample.

Normalization: Adjusting the signal amplitudes for reduction of inconsistencies between different samples.

Pre-emphasis Filtering: Emphasizing high frequencies within the voice sample for increasing clarity.

In turn, the analysis of a preprocessed signal includes examination of its wave representation in time domain. Such an analysis will provide information about the following aspects:

Intensity fluctuations of speech.

Lengths of speech segments.

Existence of any kind of pauses.

In other words, waveform analysis becomes the final stage of verification that provides information whether there is enough speech content in the input signal.

4.6.3 Frequency Analysis & MFCC Feature Extraction

There are significant properties of a speech signal that can be found in time and frequency domains. Frequency-based properties are extracted from the signal after applying transformations such as the Short-Time Fourier Transform (STFT).

The advantages of frequency domain analysis include:

Identification of distribution of speech frequencies
Identification of energy distribution across frequency bands
Detection of changes over time in the frequency patterns

Such properties will be useful in the differentiation of speech patterns based on health conditions.

Mel-frequency cepstral coefficient (MFCC) is the main feature extraction algorithm employed in the proposed model and is inspired by human hearing process. There are several stages in MFCC computation including:

Windowing (e.g. Hamming Windowing)
Short-Term Spectrum Analysis using FFT
Mapping to Mel Scale
Logarithmic Compression of Mel Spectra
Discrete Cosine Transform (DCT) of Mel Spectra

MFCC features can be represented in a mathematical format as follows:

$$\text{Equation 8 : } MFCC = DCT (\log (\text{Mel Filterbank Energy}))$$

The above equation shows how MFCC features can be calculated from the speech signal.

4.6.4 Feature Representation and Impact on Performance

Once the MFCC features have been obtained, they are presented in a specific form called a feature matrix. Namely:

Rows correspond to time frames
Columns correspond to MFCC coefficients

In this way, the model is able to study not only changes in time but also in frequency range, providing additional information from the speech.

The ordered form of the feature matrix provides a good opportunity to use transformer models which are intended to identify relationships in time sequence.

Visualization of the obtained MFCC features can be done in a heatmap or spectrogram-

like form to obtain more information about speech:

Bright areas show high energy in the specific frequency range Dark areas indicate either silence or low energy Temporal changes show speech behavior

Thus, visualization enables the validation of the correct implementation of the feature extraction and shows the distinctions in speech behavior.

The quality of the obtained features directly influences the results achieved by the model.

Thus, when the features have been well processed and properly represented, the model performs well in terms of:

Accurate predictions Ability to differentiate between the different risk categories
Resilience to noise and distortion

In conclusion, effective preprocessing and MFCC extraction allow the model to identify patterns in the speech signal that can be learned reliably and accurately by the model for diabetes risk prediction.

4.7 Model Implementation and Training Performance

This phase will entail the deployment of a transformer-based deep learning model that will classify the risk levels of diabetes using voice signals. The model utilizes the vocal biomarkers in the speech to recognize the patterns that correspond to various risk levels. The general aim of this phase is to develop an effective model that can be used to classify the risk level of the user.

The system utilizes the architecture of a transformer neural network. This type of neural network architecture is highly appropriate for modeling sequence data like speech signals. The transformer neural network architecture has several layers that allow it to extract representations of the input sequences. The input layer of the model is responsible for receiving the MFCC features extracted from the audio signal.

After embedding is done, the obtained data will be further processed using multi-head self-attention encoder layers. The latter represents the major elements of the proposed transformer model since this technique helps learn long-term dependencies in speech signals through analysis of interrelationships among various components of the input

sequence. Through attention, the model will be able to concentrate on the relevant components of the audio data, thus improving the accuracy of extracted features.

Further, the obtained features will go through feedforward neural networks that will perform transformations and processing of the learned representations. Finally, classification will be performed using a specific classification layer, which will help allocate the obtained results into the predefined risk categories. Such an approach will facilitate efficient learning and allow generalization of results based on the voice input data.

The model will be trained using a labeled dataset that consists of voices and is divided into three classes – low-risk, medium-risk, and high-risk categories. Overall, such a dataset is split into three different subsets – training set, validation set, and testing set. Training is performed using the first subset, while the second is applied during validation to avoid overfitting issues.

Hyperparameters are optimized during the training process for the successful training of the model. The training parameters used in the model are detailed as follows:

Table 2: Training Parameters

Parameter	Value
Epochs	40
Batch Size	8
Learning Rate	0.001
Optimizer	Adam
Loss Function	Cross-Entropy

Cross entropy loss is used since it can be used in multiclass classification tasks. The Adam optimizer is used in order to optimize the training process of the model. Learning rate is chosen in such a way that it does not overshoot the optimum value during training.

The learning behavior of the model is quite stable during the training process. Loss value gradually reduces during each iteration, meaning the model learns from data. After some iterations, the model becomes stable and any further training causes minor changes in loss values.

The transformer model also greatly impacts the efficiency of the learning process. The attention function speeds up the learning process by focusing on the most essential

information while ignoring less relevant information. Therefore, convergence takes place faster than in conventional deep learning models.

In conclusion, the development and learning processes for the transformer model have shown the potential of the model to effectively classify risks associated with diabetes based on voice data.

4.8 Prediction Output and Result Interpretation

Prediction output and result interpretation is the last stage in the suggested system for detecting diabetes based on voice data. Having completed audio pre-processing, MFCC feature extraction, and inference with transformer architecture model, the system produces classification results presented to the user in a clear way.

It becomes necessary to consider the module described as it is responsible for translating complex machine learning results into understandable conclusions from the perspective of medical diagnostics.

4.8.1 Prediction Generation Algorithm

When the process of extracting the feature sequence of MFCC features has been finished, the feature matrix is provided as an input to the pre-trained transformer-based model. The model then applies attention techniques to the sequential data and learns dependencies between various speech frames.

The algorithm for prediction generation consists of such steps:

Input of the sequence of MFCC feature sequence in the pre-trained model
Transformation of data in the encoder stack of the transformer model
Feature extraction from the context of speech frames
Computation of probabilities for the classes using softmax function
Outputting the class with the highest probability score

The predicted results are classified as follows:

Low risk

Medium risk

High risk

4.8.2 Probability Score Calculation

There is no class label prediction output by the model. Instead, probability scores will be predicted for each individual class.

E.g.

Low Risk → 0.18

Medium Risk → 0.27

High Risk → 0.55

Here, the high risk will be chosen as the final output due to the higher probability score than other two predictions.

Probability-based prediction gives advantages like:

Better predictability Confidence calculation Transparent decision-making

It can also help in knowing the prediction confidence level of the model.

4.8.3 Definition of Risk Categories

The system categorizes the predictions into three medically defined classes:

Low Risk Prediction indicates normal or nearly normal voice behavior No signs of any diabetic-related speech anomalies User is categorized as a healthy person Preventive care advice should be given Medium Risk Mild abnormality in voice behavior Signs that indicate some early physiological changes Not a diagnosed condition, but a warning one Consultation of medical doctors advised High Risk Strongly correlated with diabetic speech behaviors Deviation in voice behavior from normal cases Medical examination is an emergency requirement Most crucial category within the system.

4.8.4 User Interface Display of Prediction Outcome

Prediction outcomes are presented in an easy and interactive way by the system.

The prediction outcome display involves:

Label of risk group Highlighting of result portion Optional use of visual elements (such as coloring and alerts) Easy-to-understand interpretation of health state

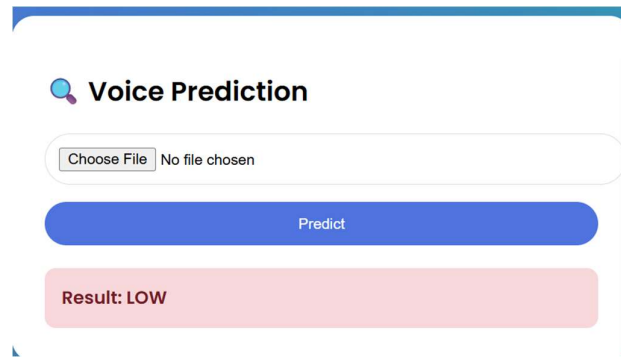


Fig 4.8 1 Output for low risk prediction

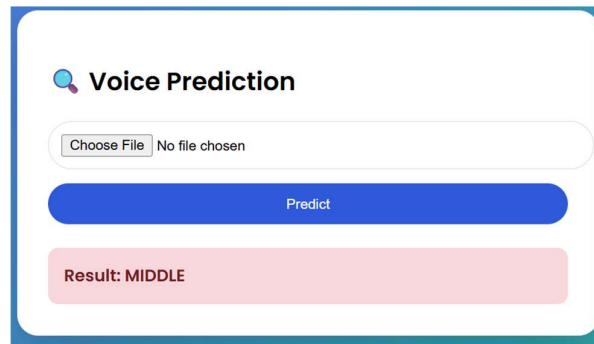


Fig 4.8 2 Output for medium risk prediction

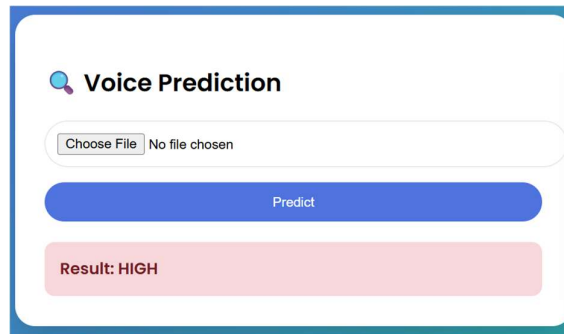


Fig 4.8 3 Output for high risk prediction

As can be seen from these illustrations, the output is presented for all three risk groups separately.

4.8.5 Health Advice & User-Centric Output Design

In addition to offering the prediction results, the proposed solution puts emphasis on providing useful information to users. As the program is aimed at general users without any technical knowledge or medical background, the output format is made clear and easy to understand.

Depending on the risk factor (Low, Medium, or High), the system offers general recommendations and guidelines which will help users make decisions based on their situation. This approach adds value to the program, making it more convenient to use and apply in practice.

The following health recommendations are generated by the system:

Balancing and maintaining a proper diet to control the levels of glucose
Getting physically active and exercising regularly to improve overall health condition
Going through health check-ups to detect the disease in advance
Seeking consultation from doctors especially when the risk factor is medium or high

It should be noted that the above health recommendations are generally provided and do not represent professional medical advice.

Along with giving health recommendations, simplicity in the output is encouraged to facilitate understanding by the users. Rather than showing complicated technical information, the following information is provided in the output:

Risk Category: Low, Medium, High

What does this mean for health?

Interpretation of the prediction: What action should the user take?

This kind of presentation guarantees that the output is understood without any difficulty and confusion by the users.

4.8.6 Prediction Performance and Confidence Analysis

The system is intended to be capable of delivering predictions in almost real-time, enabling users to get instant results following their data entry. Predictions are made according to a pipeline of actions carried out efficiently:

Processing of the entered audio stream
Feature extraction (MFCC)
Model inference with the trained transformer
Prediction results generation and presentation

It is important to note that various parameters have an impact on the accuracy of predictions made by the proposed approach.

Audio quality: Good audio quality helps achieve more accurate predictions
Background noise: It can interfere with the feature extraction process
Speaker variability: Speaker

accent, tone, and pitch variation affects the model performance Quality of recording equipment: Microphones sensitivity influences the results obtained Diversity of training data: It helps increase the accuracy of predictions

In order to solve these issues, some preprocessing procedures (such as filtering) were used.

Another critical feature is the confidence of predictions, which gives an understanding of the reliability of the result provided by the model.

For instance:

Confident prediction: High Risk - 0.85

Unconfident prediction: Medium Risk - 0.40, High Risk - 0.42

Higher probability reflects the greater confidence in the result, whereas close probabilities denote lack of confidence. In this situation, the class that has a higher probability value will be selected; however, it should be noted that further medical examination will still be required.

Such interpretation of results can help users determine their credibility and make informed decisions. This aspect also explains why the model is to be used only as an auxiliary tool and not a diagnostic one.

4.9 Chatbot Interaction & Response Analysis

The chatbot is developed as an interactive module of the proposed system to offer instant assistance to the user. In the process of implementing the system, the chatbot is incorporated into the web-based interface of the proposed system, thereby enabling users to interact directly with the system through inquiries concerning diabetes information, prediction results, and health advice.

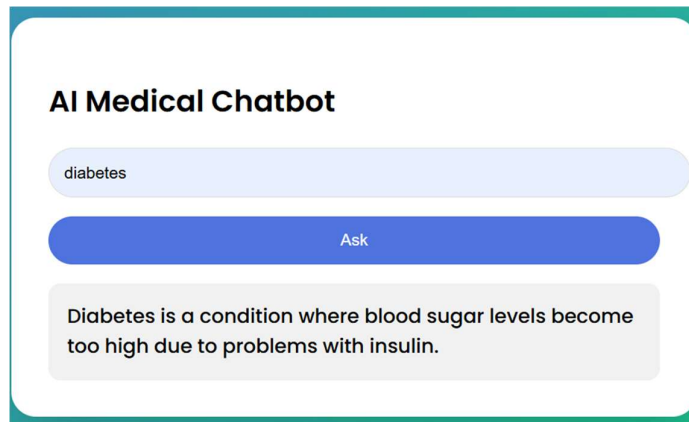


Fig 4.9 1 Chatbot interaction interface

As shown above, the chatbot offers a very convenient interface for the user to interact with the proposed system in a more efficient manner.

4.9.1 API Key Integration and Validation

While implementing, the chatbot connects itself to the external language model via an API key. This API key serves as the critical factor in allowing the chatbot to generate intelligent responses dynamically.

API Keys

Key Status: Active

[+ Create key](#)

Name	ID	Key	Status	Created	Created By	Last Used
Legacy API Key	key_CYwhNJGcDJSsQX6dT3zPy	9e11	Active	3/11/2026, 11:05 PM	dasariakanksha7@gmail.com (You)	3/12/2026, 12:29 AM
voice	key_CYwiqSSMYXiNqcE63oE8G	mfgw	Active	3/11/2026, 11:24 PM	dasariakanksha7@gmail.com (You)	3/11/2026, 11:25 PM
aka	key_CYyVTjUAff7iwVvaNXyYN	DbkY	Active	3/12/2026, 9:50 PM	dasariakanksha7@gmail.com (You)	N/A

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Fig 4.9 2 API key setup and validation window

In the implemented system:

API key is kept safely stored and validated at runtime

Valid API key means communication with external APIs will be enabled

Invalid or no API key means fallback operation will be performed automatically

This way, even in scenarios where the external API services are not available, the chatbot will continue to work.

4.9.2 Dynamic Response Generation (API Mode)

Once the validation process is successful for the API key, the chatbot is enabled to produce dynamic as well as context-sensitive responses via an external language modelling API service.

In this mode of operation:

The user query is passed to the external API service. The API evaluates the input and returns intelligent responses. The responses are shown on the chatbot screen.

The API-based system thus allows handling complex user inputs including questions such as:

“What is the first sign of diabetes?” “What is the working of voice detection?” “What to do when there is high risk?”

4.9.3 Rule-Based Response Generation (Fallback)

If the validation process fails due to the expiry or malfunction of the API key, then the system shifts to the rule-based fallback option.

In this mode of execution:

Pre-defined pairs of questions and answers are maintained in the system. Keyword recognition is used to recognize the query. Appropriate pre-defined responses are shown instantly.

This ensures:

Constant availability of the chatbot. No system failure due to problems with the API service. Functionality even in the offline mode.

4.9.4 User Interaction & Real-time Testing of Queries

Several user queries were carried out during testing and implementation phases to check the chatbot efficiency. The following were some successful queries answered by the bot:

Queries on diabetic health issues Usage queries Explanations for prediction outcomes
General input queries

The bot answered the above questions quickly and meaningfully showing that the bot had been successfully incorporated into the front-end interface.

4.9.5 Interpretation of Chatbot Responses

The responses received from the bot have been kept:

Simple and easy to interpret Brevity to the point Information rich but not technically loaded

The above qualities have ensured ease in interaction with the system even for non-technical users.

4.9.6 Observation of Performance in Implementation

In the implemented system, the chatbot has the following advantages:

Rapid response time in both API and rules systems Effortless integration with the web interface Stable performance on multiple queries Rapid response generation without causing any system delay

Combining an API and a rule-based system makes the process more reliable and easier to use.

4.9.7 Significance in Implemented System

In the implemented system, the chatbot will be beneficial in:

Facilitating conversation between the user and the system Helping the user understand predictions made by the system Creating better user experience Making the system more applicable to real-life situations

Overall, the chatbot makes the system more interactive than just a tool to make predictions.

4.10 System Performance and Accuracy Analysis

This analysis stage assesses the efficiency of the developed model regarding prediction accuracy and other essential parameters such as response speed and reliability. This step

is necessary to ensure that the implemented model is efficient enough for practical use according to the specified criteria. The analysis is performed when all modules, such as audio input, MFCC feature extraction, transformer classification, and chatbot interaction, have been successfully developed.

Table 3: Performance Comparison of Existing Methods

S.No	Year	Dataset	Method	Accuracy (%)
1	2019	IDRiD	CNN + Handcrafted Features	90.70
2	2020	IDRiD	CANet	92.60
3	2020	IDRiD MESSIDOR	CNN	90.29 90.89
4	2020	IDRiD	RSNet	86.33
5	2020	Kaggle	CNN	81.00
6	2021	IDRiD	Fine KNN	94.00
7	2022	IDRiD	XGBOOST	63.00
8	2025	IDRiD	SVM-RBF	64.00
Proposed Work	2026	IDRiD	Transformer	94.87

4.10.1 Performance of Model & Analysis of Training

The performance of the voice-based diabetes detecting system proposed can be determined in terms of classification accuracy and properties of its training. Classification accuracy means the rate of correctness in predictions and is a good criterion for evaluating the quality of the model. It is possible to say that the system proposed provides relatively high accuracy because it is capable of examining complicated speech signals with the help of MFCC features and transformer architecture.

In turn, the feature extraction procedure based on MFCC ensures taking into account the important frequency-related aspects of the human voice, which is essential in order to detect changes related to diabetes. Moreover, the examination of such data is conducted by means of transformer architecture, which is suitable for dealing with long-range dependencies that can be present in voice data. At last, the attention mechanism allows focusing only on relevant parts of the data.

Training of the model is conducted via cross entropy loss which is appropriate for multiclass classification tasks. The model learns during the training as the loss value drops gradually with every training iteration. After a certain number of iterations, the model stabilizes itself and the loss value starts converging. Moreover, the validation

loss is quite close to the training loss which suggests that there is no overfitting issue in the model. Therefore, the mentioned training process proves that the model has learned the patterns in the dataset.

In addition to this, the system exhibits excellent performance while compared with other models including support vector machines, random forest, and logistic regression. In particular, the presented approach differs from these algorithms due to the possibility of processing sequential data and building context within the speech signals. This makes the model superior to other machine learning approaches.

4.10.2 System Efficiency, Reliability, & Overall Evaluation

Besides the accuracy of predictions, the efficiency of the system is measured with regard to its reaction speed and ability to process data in real-time. In fact, the application is able to provide near real-time predictions, thus allowing the user to get results smoothly. The efficient audio preprocessing stage includes both normalization and elimination of noise from the sound samples. Furthermore, the system quickly processes data during the MFCC features extraction stage. At the same time, the trained transformer model allows predicting the emotions of people quickly.

System Reliability and Stability

The reliability and stability of the system are evaluated when considering various input parameters. The test results prove that the application provides accurate predictions of emotions for a person speaking in various tones, pitches, accents, and even with background noises. These stable test results are reached due to proper preprocessing, namely noise reduction and normalization, and the reliable operation of the transformer model.

Despite this, there are some limitations noted during the testing phase. For instance, variations in accuracy may occur in cases where there is inconsistency in the features of the speaker and environmental conditions. Variations in noise levels and input devices can also be responsible for poor input data.

As mentioned above, the system assessment proves to be satisfactory because of numerous positive characteristics exhibited by the proposed approach. Some of the most important qualities include accuracy, reliability, and rapid performance of the model.

In addition to this, the system can be easily integrated with the chatbot module and others.

To conclude, the performance metrics clearly indicate that the proposed system is highly efficient and effective in terms of voice-based detection of diabetes.

4.11 Discussion of The Result

Discussion of the result provides a comprehensive evaluation of the results of implementation of the proposed diabetes diagnosis based on the use of voice signals. The part is dedicated to the assessment of efficiency of the application of such a method as well as its analysis in order to evaluate the importance of the outcome of using the transformer-based model, MFCC feature extraction and the chatbot. The main findings prove that the suggested system has the ability to analyze voice signals and categorize the results of such analysis in terms of diabetes risk level.

4.11.1 Efficiency of Voice-Based Diagnosis

The results obtained proved that it is possible to use information about the voice characteristics to predict diabetes risks. With the help of MFCC feature extraction, a transformer can learn useful representations from the speech signal. The following conclusions can be made from the findings:

It is possible to map voice signals to diabetes risk groups efficiently

The usage of MFCC significantly improves classification results

Transformer architecture positively influences pattern recognition capabilities

4.11.2 Analysis of Prediction Outcomes

The output of predictions produced by the system includes:

Low Risk, Medium Risk, High Risk

From the output we can observe that:

Cases where the risk is low are detected accurately in most of the scenarios

In some cases, medium risk may coincide with low risk as they have similar voice features

The cases of higher risks are distinctly identified as there are strong differences in the features

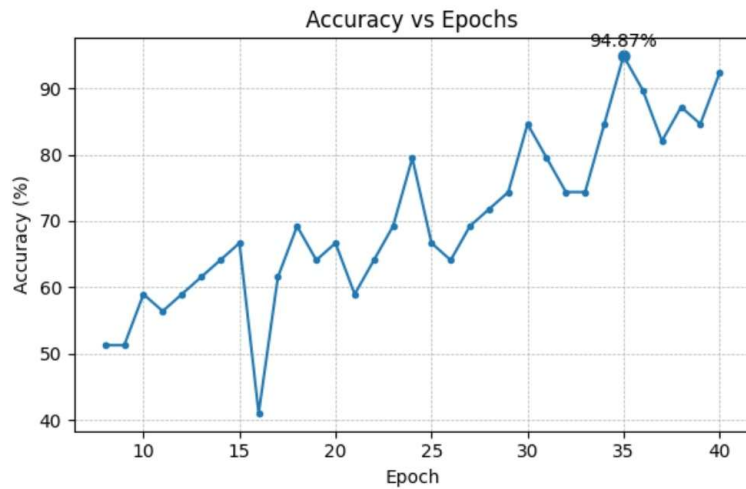


Fig 4.11.2 1 Performance and accuracy analysis

4.11.3 Performance, Usability, and Limitations

Performance of the developed system shows great efficiency of using the latest methods for features extraction, application of neural network modeling, and incorporating the interactive elements. One of the most important factors behind success of the suggested system is the use of Mel Frequency Cepstral Coefficients and a transformer model that allows for conducting proper analysis of voice data for detecting the risk of diabetes.

MFCCs are responsible for identifying valuable features that reflect the spectrum of speech signals. In particular, these coefficients describe the frequency-related information of human voice, thus, becoming an extremely useful way to conduct machine learning. The focus on perceptual frequency aspects allows for detecting all valuable patterns within speech data that can be used to diagnose diabetes.

The transformer model further improves the performance of the system by using the sequentiality of voice data. As opposed to standard machine learning algorithms that can predict only based on short-term memory, the transformer is able to capture the relationships between various aspects of input data over time. With a self-attention mechanism implemented in the model, the machine learns to pay attention to only relevant parts of the voice recordings, thus increasing the accuracy of prediction. Combining the two techniques leads to superior results when predicting diabetes from voice recordings.

Apart from improving prediction accuracy, usability of the system was increased by the inclusion of the chatbot module. Acting as a digital assistant to users, the chatbot contributes to improved user experience as it helps interpret the prediction results. Apart from helping to make sense of the prediction results, the chatbot is useful in delivering other information about diabetes, like symptoms, risk factors, and preventative measures for developing the condition.

In spite of the success of the approach, some disadvantages have been noted during the testing and evaluation process. First, the classification algorithm sometimes produces inaccurate predictions in the differentiation between the low-risk and medium-risk classes. This is primarily caused by the intersection of feature values in the speech data. Another problem associated with the proposed model concerns its sensitivity to environmental noises that might impact the quality of the received signal. Besides, differences in voice parameters (e.g., accent, tone, or pitch) might influence prediction outcomes in an adverse manner. Moreover, the system has another significant disadvantage related to its reliance on the initial training dataset.

This aspect requires further attention because the performance of the model depends on the diversity and complexity of training examples used.

The overall findings from the analysis of the results indicate that the suggested system meets the goals set out to be accomplished. It manages to classify diabetes risk levels based on voice data in a non-invasive manner and with reasonable speed for real-time use. The suggested system operates effectively in controlled settings and manages to combine deep learning and interactive capabilities. Audio signal processing, transformer model training, and chatbot interaction make up a unique combination for diabetes risk predictio

CHAPTER-5

CONCLUSION

5.1 Conclusion

The proposed voice-based diabetes diagnosis system incorporates an innovative and intelligent healthcare solution that helps detect diabetes risk at the onset and in a non-invasive way. Thanks to the effective implementation of advanced technologies such as machine learning, deep learning, and the web, this system meets the growing needs of contemporary healthcare settings regarding easy, accurate, and fully automated diagnostics. Early detection of the mentioned disease and its related complications by analyzing voice features represents the major goal of the current research.

In the course of this study, MFCCs (Mel Frequency Cepstral Coefficients) were chosen as feature extraction technology to analyze voice signals. This method enables the extraction of important acoustic features, which are critical for detecting subtle differences between healthy and diabetic voices. The obtained features are further analyzed using a deep learning algorithm based on the Transformer architecture. When compared to other machine learning algorithms, the mentioned solution demonstrates better results in terms of learning ability, generalization, and prediction quality.

This model categorizes users' inputs in the form of different diabetic risk groups such as low, moderate, and high diabetes risk levels, helping users interpret their current health status easily. Implementation of the Django web application helps create an easy flow of conversation between the user and system where users can make their voices heard and receive instant feedback through an easily usable interface.

Moreover, the chatbot module has increased the functionality of the system, which is also now capable of delivering interactive replies related to the awareness of diabetes, its symptoms, and other general health concerns.

The experimental analysis proves that the suggested system can provide better accuracy than the traditional ML models while being reliable and easy to use due to fast real-time

processing ability.

On the whole, this study illustrates the enormous possibilities of artificial intelligence when it comes to improving the current healthcare systems into intelligent and cost-effective alternatives. The system's ability to eliminate the need for invasive procedures and help identify patients at risk of developing diseases earlier is important in terms of preventive health care.

In sum, the proposed system has successfully achieved its set goals of offering a non-invasive and automatic early diabetes prediction solution that could be a good basis for future work in this direction.

5.2 Future Scope

The presented solution represents a few ways that should be explored to enhance the effectiveness of intelligent health care systems. While the solution is rather efficient today, the scope of possible improvements remains wide.

Firstly, one of the critical areas that need to be enhanced refers to integrating the system with wearables like smartwatches and biosensors that allow collecting health data in real time. Integrating information about pulse and other indicators along with voice data can provide an opportunity to predict the patient's state more accurately.

Secondly, another enhancement that can be applied relates to expanding the scope of voice samples collected during data processing to cover a wide range of cases. Increasing the volume of available data may allow training a more generalized and robust model and hence make the system more efficient when applied in practice.

The proposed system can be deployed to cloud platforms for providing better performance and making it accessible worldwide.

Other optimizations can be made regarding further enhancing the efficiency of the model used by applying advanced machine learning approaches.

The chatbot module can be enhanced to become an intelligent AI-based health assistant that is able to provide tailored advice, dietary plans, and ongoing health check suggestions for users.

Moreover, the application can be further developed to identify various types of chronic illnesses including cardiovascular and respiratory ailments through voice-based

biomarkers. This will enable the creation of an AI-based multi-disease healthcare solution.

Furthermore, incorporation into mobile apps can greatly enhance convenience, making it possible for users to conduct health checks at any time from anywhere by using their mobile phones.

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CO-PO MAPPING

Course: Project (22ECC35)

Course Outcomes:

1. Develop a voice-based system for early diabetes detection using speech signal processing and machine learning.
2. Integrate real-time voice input processing and cloud-based model inference.
3. Analyze and interpret voice biomarkers using transformer-based deep learning models.

S.No.	Description of Learning Outcomes	CO	PO	Bloom's Level
1	Studied and analyzed the significance of non-invasive diabetes detection using voice signals and AI techniques.	CO1	PO1, PO2, PO4	Understand, Analyze
2	Collected and preprocessed voice data (feature extraction like MFCC, pitch, tone) for model training.	CO2	PO1, PO5	Apply, Understand
3	Designed and implemented a Transformer-based model for voice-based diabetes prediction.	CO2	PO3, PO5	Apply, Analyze
4	Developed a system integrating voice input, model inference, and prediction output using cloud or local deployment.	CO2	PO3, PO6	Analyze, Apply
5	Built an intelligent chatbot support system for user interaction and diabetes awareness.	CO3	PO3, PO5, PO6	Apply, Create

6	Evaluated system performance using metrics like accuracy, confusion matrix, and ROC curve.	CO4	PO9, PO10	Create, Evaluate
7	Documented and presented the project with technical clarity, innovation, and effective communication.	CO5	PO8, PO9, PO11	Apply, Create

SDG MAPPING

Alignment with United Nations Sustainable Development Goals (SDGs)

The project titled **“Voice-Based Early Diabetes Diagnosis Using a Transformer Model with Intelligent Chat Support”** supports several United Nations Sustainable Development Goals (SDGs) that focus on improving healthcare accessibility, promoting awareness, and leveraging advanced technologies for early disease detection. The system uses voice analysis and artificial intelligence to provide a non-invasive, cost-effective solution for diabetes diagnosis. Additionally, the integration of an intelligent chatbot enhances user interaction and awareness, making healthcare support more accessible to diverse populations. The alignment is outlined below:

SDG	Goal Title	Project Contribution
SDG 3	Good Health and Well-being	Enables early detection of diabetes using non-invasive voice analysis, improving preventive healthcare and reducing health risks.
SDG 4	Quality Education	Provides awareness about diabetes through intelligent chatbot support and promotes health education among users.
SDG 9	Industry, Innovation and Infrastructure	Utilizes advanced AI (Transformer models) and speech processing technologies to build innovative healthcare solutions.



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